

VITEBSK, Belarus

WMO: 266660

Lat: 55.127N

Lon: 30.350E

Elev: 208

StdP: 98.85

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-18.2	-24.6	0.4	-21.2	-21.3	0.6	-17.4	9.6	-1.1	8.5	-2.5	1.6	20	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	28.9	20.0	27.0	19.0	25.2	18.0	20.9	27.0	19.9	25.2	19.0	23.9	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.7	17.9	13.2	22.5	17.0	12.5	21.7	61.2	27.3	57.8	25.4	54.8	23.9	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.4	-25.7	31.4	3.9	2.2	-28.5	33.0	-30.7	34.3	-32.9	35.5	-35.7	37.2		
(5)			-25.9	22.5	3.8	0.9	-28.6	23.1	-30.9	23.6	-33.0	24.1	-35.8	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-5.6	-5.1	-0.2	7.3	13.1	16.7	19.2	17.7	12.2	6.2	0.9	-3.5	(6)
(7)		DBStd	9.92	6.20	6.39	4.55	4.59	4.59	3.42	3.09	3.46	3.52	4.05	4.79	5.81	(7)
(8)		HDD10.0	2191	483	423	317	107	21	1	0	0	18	129	274	418	(8)
(9)		HDD18.3	4387	741	656	575	331	171	72	26	54	184	376	524	676	(9)
(10)		CDD10.0	964	0	0	1	26	118	201	284	238	86	12	0	0	(10)
(11)		CDD18.3	118	0	0	0	0	9	22	52	33	1	0	0	0	(11)
(12)		CDH23.3	859	0	0	0	2	74	125	372	274	12	0	0	0	(12)
(13)		CDH26.7	194	0	0	0	0	12	14	87	81	1	0	0	0	(13)
(14)	Wind	WSAvg	3.0	3.1	3.0	3.2	3.1	2.9	2.8	2.6	2.7	3.0	3.2	3.3	(14)	
(15)	Precipitation	PrecAvg	734	55	49	44	39	64	78	89	77	65	66	58	51	(15)
(16)		PrecMax	929	107	84	99	86	115	148	208	290	139	152	114	90	(16)
(17)		PrecMin	553	9	18	12	9	19	20	20	8	18	13	0	21	(17)
(18)		PrecStd	101	22	18	18	21	28	32	49	57	35	36	25	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	6.1	13.8	22.9	28.1	28.7	31.2	32.9	25.0	19.0	11.6	8.1	(19)
(20)			MCWB	4.6	4.3	7.6	13.2	18.7	19.8	20.7	20.8	17.3	13.1	10.2	7.2	(20)
(21)		2%	DB	3.5	3.9	10.2	19.9	25.4	26.5	29.1	28.4	22.0	15.0	9.6	6.2	(21)
(22)			MCWB	2.7	2.5	5.2	11.6	17.5	18.5	20.4	19.5	15.6	12.0	8.5	5.5	(22)
(23)		5%	DB	2.3	2.8	7.7	17.2	23.0	24.8	27.1	26.2	19.7	13.2	7.9	4.5	(23)
(24)			MCWB	1.6	1.7	4.2	10.3	15.6	17.8	19.6	18.6	14.7	11.1	6.8	3.9	(24)
(25)		10%	DB	1.2	1.8	5.6	15.0	20.8	23.1	25.2	24.0	17.9	11.8	6.7	2.8	(25)
(26)			MCWB	0.6	1.0	3.1	9.3	14.3	16.8	18.5	17.7	13.7	10.1	5.9	2.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	4.6	8.2	14.2	19.9	21.0	22.6	21.6	18.1	14.1	10.6	7.4	(27)
(28)			MCDWB	5.2	5.8	12.6	21.0	26.2	26.3	28.8	29.8	23.6	16.8	11.4	7.8	(28)
(29)		2%	WB	2.7	2.8	6.2	12.4	18.2	19.7	21.3	20.5	16.3	12.8	8.7	5.6	(29)
(30)			MCDWB	3.2	3.5	9.3	17.6	24.0	24.5	27.2	26.9	20.0	14.5	9.4	6.0	(30)
(31)		5%	WB	1.7	1.9	4.6	11.1	16.6	18.7	20.2	19.5	15.2	11.4	7.0	3.8	(31)
(32)			MCDWB	2.2	2.5	6.7	15.7	21.9	23.2	25.6	24.8	18.5	12.8	7.6	4.5	(32)
(33)		10%	WB	0.7	1.0	3.3	9.9	15.2	17.6	19.2	18.4	14.2	10.2	5.9	2.1	(33)
(34)			MCDWB	1.1	1.6	5.3	14.3	19.6	21.8	24.2	22.8	17.0	11.7	6.5	2.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	5.6	6.6	8.6	9.4	8.7	9.1	9.3	8.2	5.7	3.9	3.8	(35)
(36)			MCDBR	3.5	4.2	9.0	12.0	12.0	10.9	11.4	12.5	11.1	7.6	4.9	3.6	(36)
(37)			MCWBR	3.5	3.4	5.6	5.9	5.6	5.1	4.9	5.3	5.9	5.2	4.1	3.3	(37)
(38)		5% WB	MCDWB	3.2	3.6	7.6	10.5	11.2	9.5	10.5	10.7	9.8	6.5	4.3	3.6	(38)
(39)			MCWBR	3.3	3.2	5.2	5.7	5.6	5.1	5.0	5.0	5.8	4.9	4.0	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.320	0.348	0.401	0.390	0.374	0.401	0.410	0.367	0.347	0.327	0.289	(40)	
(41)		taud	2.154	2.108	2.156	2.202	2.312	2.379	2.320	2.287	2.378	2.394	2.253	2.172	(41)	
(42)		Ebn at Noon	622	731	809	817	852	871	837	801	791	714	572	528	(42)	
(43)		Edh at Noon	62	93	113	125	119	112	117	114	91	71	57	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.61	1.50	2.90	3.99	5.18	5.60	5.40	4.35	2.86	1.35	0.48	0.36	(44)	
(45)		RadStd	0.14	0.25	0.39	0.47	0.48	0.52	0.50	0.45	0.24	0.28	0.09	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page